

Decision Tree Practice Question:

Decision Tree Question (Different Dataset)

You are analyzing whether a customer will **buy a product** based on three features:

ID	Age Group	Income Level	Previous Purchase	Will Buy
1	Young	High	No	No
2	Young	High	No	No
3	Middle	High	No	Yes
4	Senior	Medium	No	Yes
5	Senior	Low	Yes	Yes
6	Senior	Low	Yes	No
7	Middle	Low	Yes	Yes
8	Young	Medium	No	No
9	Young	Low	Yes	Yes
10	Senior	Medium	Yes	Yes
11	Young	Medium	Yes	Yes
12	Middle	Medium	No	Yes
13	Middle	High	Yes	Yes
14	Senior	Medium	 No	No

Tasks to Solve:

1. Calculate the Entropy of the full dataset for the target Will Buy.
2. For each feature (**Age Group**, **Income Level**, **Previous Purchase**), calculate the Information Gain when used for splitting.
3. Which feature gives the highest Information Gain? (**=> Choose this as the root node.**)
4. Draw the first split of your Decision Tree based on the best feature.